Amit Rotem

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Programming: C++20, Python, CUDA, MPI, OpenMP

Interests: high performance scientific computing, mathematical modeling, partial differential equations, machine learning.

Education

Virginia Tech, Blacksburg, VA — *Ph.D. Mathematics, Expected 2026*
GPA: 3.9

Colorado School of Mines, Golden, CO — *M.S. Computational & Applied Mathematics, 2021*
GPA: 4.0

Colorado School of Mines, Golden, CO — *B.S. Applied Mathematics & Statistics, 2020*
GPA: 3.6

Technical Skills

Languages	C++20, Python (NumPy, pandas, scikit-learn, TensorFlow, PyTorch), MATLAB, GoLang
Computation	CUDA, MPI, OpenMP
Numerics	finite elements, iterative methods, domain decomposition, low-rank, machine learning
Tools	Linux, Git, CMake, profiling/optimization, distributed computing systems

Projects & Experience

Lawrence Livermore National Laboratory — Computing Scholar — Summer 2024

– Devised matrix-free methods (C++/CUDA) targeting high dimensional physics simulations.
– Implemented templated GPU kernels supporting configurable memory/thread layouts.
– Code: https://github.com/GenDiL/GenDiL

High-Performance Discontinuous Galerkin (DG) in MFEM Library — 2023–Present

– Integrated interior penalty DG in 2D/3D with GPU-optimized matrix-free algorithms.
– Achieved substantial speedups relative to existing matrix-based MFEM implementations.
– Code: https://github.com/mfem/mfem

Ph.D. Research — Virginia Tech — 2021–Present

– Developed theory and practical implementations of the WaveHoltz iterative Helmholtz solver.
– Extended multiscale variant for homogenization.
– Built scalable domain-decomposition and multigrid accelerated variants.
– Formulated fast low-rank reduced order models for the Navier Stokes equations.

CASERM (NSF IUCRC) — 2019–2021

– Built ML pipelines for hyperspectral geological imaging: segmentation, clustering, classification.
– Constructed predictive models in TensorFlow and scikit-learn.

Selected Publications

– Rotem A., Runborg O., Appelö D. *Convergence of the Semi-Discrete WaveHoltz Iteration*. *Journal of Computational Physics,* 2026.
– Rotem A. *Efficient Domain Decomposition for the Helmholtz Equation on GPUs.* *Preprint on ArXiv,* 2026.
– Rotem A., Runborg O., Appelö D. *The WaveHoltz Heterogeneous Multiscale Method.* *Preprint on ArXiv,* 2026.
– Carter S., Rotem A., Walker S. *A Domain Decomposition Approach for Nematic Liquid Crystal Models*. *J. Non-Newtonian Fluid Mechanics*, 2020.
– Rotem A. et al. *Interpretation of Hyperspectral SWIR Core Scanning Data Using ML*. *Geosciences*, 2023.

Awards

– Best Poster at SIAM Central States, 2024.
– Professor Everett Award (mathematical contributions to geological sciences) at Colorado School of Mines, 2021.